AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 3)

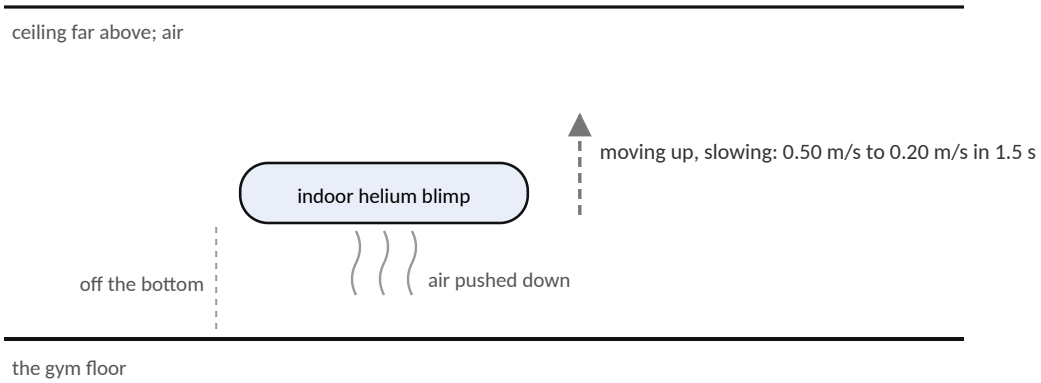
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

For all questions:

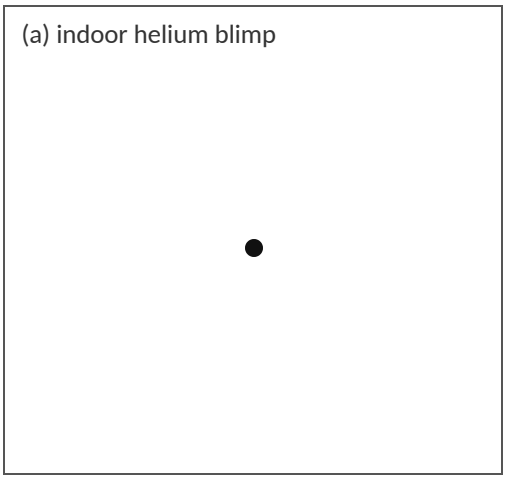
- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) A 0.450 kg indoor helium blimp has lifted off the gym floor. Its propellers push air downward, so the air pushes it upward with a thrust of 0.60 N. After it leaves the bottom, still moving upward, its speed drops from 0.50 m/s to 0.20 m/s in 1.5 s.



Assumptions for this practice problem: *air drag on the indoor helium blimp is negligible; its acceleration is constant; g = 9.8 m/s².*

(a) Draw the free-body diagram of the indoor helium blimp after it has left the bottom.

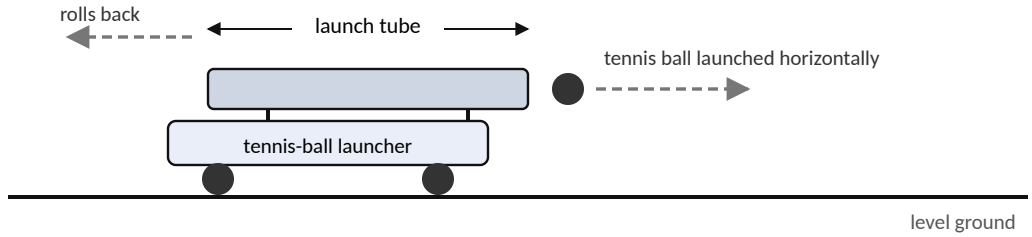


(b) Calculate the indoor helium blimp's acceleration, magnitude and direction.

(c) With up as positive, write Newton's second law for the indoor helium blimp and solve it for the buoyant force in letters.

(d) Calculate the magnitude of the buoyant force on the indoor helium blimp.

2. (25 points) A tennis-ball launcher strapped to a skateboard launches a tennis ball horizontally to the right. Find the tennis-ball launcher's acceleration while the tennis ball is in the tube.



tennis ball mass	57.0 g
mass of the launcher and skateboard	3.00 kg
launch tube length	32.6 cm
distance the ball moves relative to the ground while in the tube	32.0 cm
exit speed, relative to the ground	16.0 m/s

The ball moves less than the tube length relative to the ground, because the launcher rolls back while the ball is in the tube.

Assumptions for this practice problem: *the tennis-ball launcher starts at rest on level ground and rolls freely (ignore friction and air resistance); the tennis ball starts from rest; its acceleration in the tube is constant; the push between the tennis ball and the tennis-ball launcher is the only horizontal interaction.*

(a) Draw the free-body diagram of the tennis-ball launcher while the tennis ball is in the tube.

(a) tennis-ball launcher

(b) Calculate the tennis ball's acceleration relative to the ground while the tennis ball is in the tube.

(c) Calculate the magnitude of the force the tennis-ball launcher exerts on the tennis ball, and the magnitude of the force the tennis ball exerts on the tennis-ball launcher.

(d) Calculate the tennis-ball launcher's acceleration, magnitude and direction.

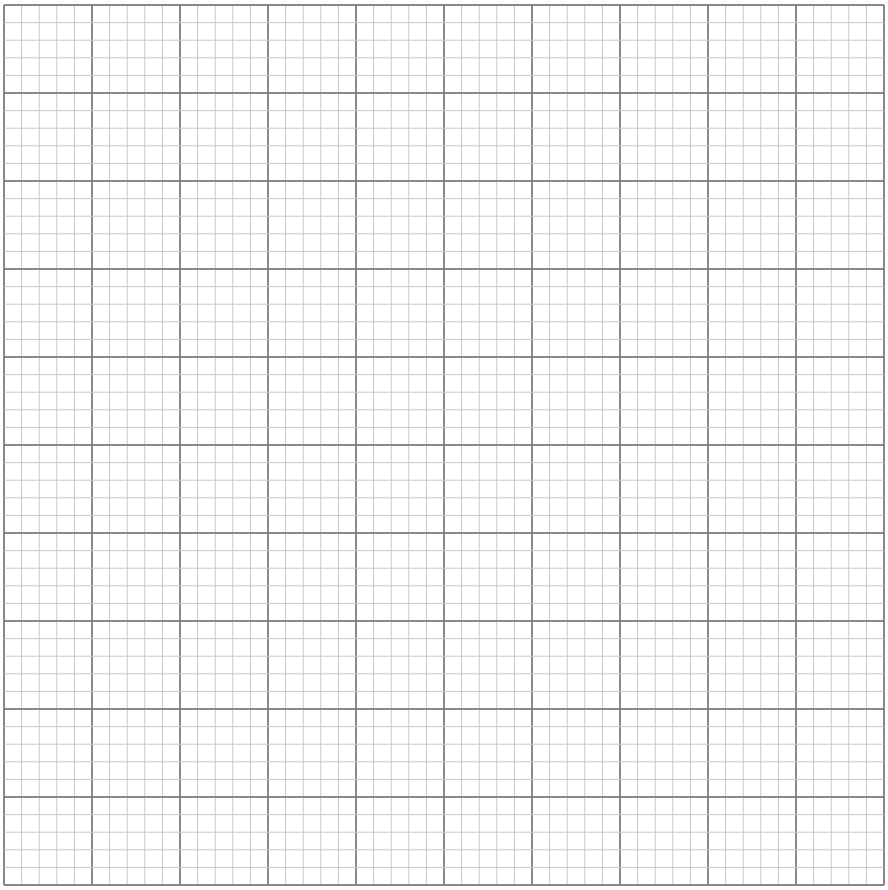
3. (25 points) Students measure the drag force F_D on a kite in a wind tunnel at several air speeds v. The drag force is modeled by $F_D = Kv^2$, where K is a constant.

Speed v (m/s)	**Drag force F_D (N)**	
2.00	1.62	
4.00	6.40	
6.00	14.9	
8.00	28.1	
10.0	42.5	
12.0	60.7	

(a) Using $F_D = Kv^2$, state the quantity you will plot on the vertical axis and the quantity you will plot on the horizontal axis so the data form a straight line, each with its units. State what the slope equals.

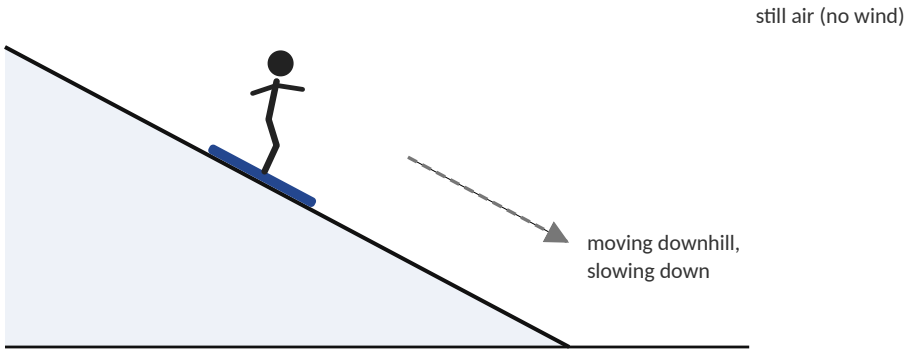
(b) Use the blank column in the table to calculate the values you will plot on the horizontal axis.

(c) Plot the data on the grid. Label each axis with the quantity and units, choose a scale, and draw the best-fit line.



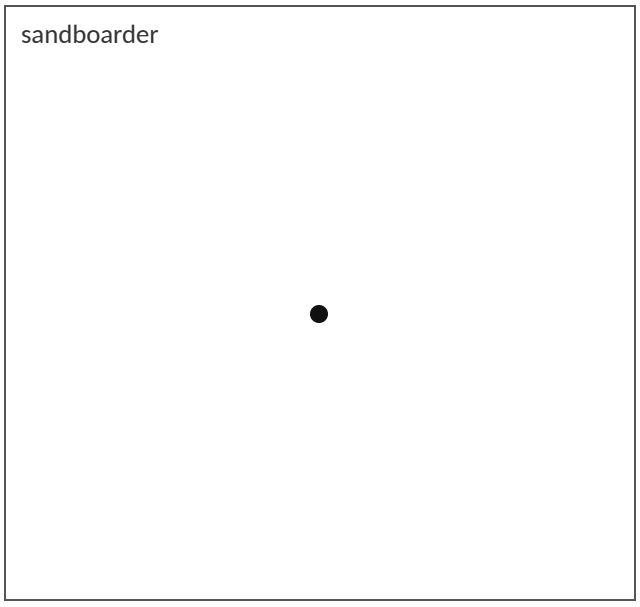
(d) Calculate the slope of your best-fit line using two well-separated points on the line. Then give K with its units.

4. (25 points) A sandboarder slides down a sand dune. In the picture the sandboarder moves down the slope to the right and is slowing down. Friction from the sand and drag from the air are **not** negligible. No angle is given.



Assumptions for this practice problem: *the air is still (no wind), so the air moves past the sandboarder opposite its motion.*

Draw and label the free-body diagram of the sandboarder. Draw the forces themselves, not components. Each force is one arrow that starts on the dot and points in the direction the force acts. No calculations.



Answer Key – Practice Test, Version 3 – Introductory Dynamics

Worked step by step: diagram, symbolic equations and algebra first, then numbers with units. g = 9.8 m/s² does not limit significant figures; answers match the significant figures of the givens.

Numerical problems: diagram first, then a = ΣF/m (equation sheet), written a = F_{net}/m. Replace F_{net} with the actual forces and their signs, swap F_g = mg, solve in letters, then plug in numbers with units.
Kinematics: #1 $v = v_0 + at$ · #2 $\Delta x = v_0 t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Thrust and buoyancy (25 points)

indoor helium blimp, off the bottom



Step 1 **(a) Forces on the indoor helium blimp after it leaves the bottom.**

F_g down (Earth). Thrust F_t up (the air it pushes down pushes back up on it). Buoyancy F_b up (the surrounding air).

No normal force: it is no longer touching the bottom. Acceleration is down (slowing while moving up), so $F_t + F_b$ is shorter than F_g.

Step 2 **(b) Acceleration from the motion data.**

Two speeds and a time: use #1. Up is +, and both velocities are upward.

$$v = v_0 + at \rightarrow a = \frac{v - v_0}{t} = \frac{0.20\ \text{m/s} - 0.50\ \text{m/s}}{1.5\ \text{s}} = -0.20\ \text{m/s}^2 \rightarrow \boxed{\textbf{0.20 m/s², downward}}$$

It is moving up but slowing down, so the acceleration points down.

Step 3 **(c) Newton's second law, up is +, letters first.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_b + F_t - F_g}{m} \rightarrow F_b + F_t - mg = ma$$

$$\boxed{F_b = m(g + a) - F_t}$$

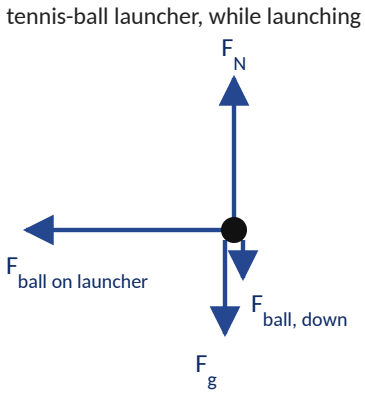
Step 4 **(d) Now the numbers, units in every parenthesis.**

$F_b = (0.450\ \text{kg})(9.8\ \text{m/s}^2 + (-0.20\ \text{m/s}^2)) - 0.60\ \text{N} = (0.450\ \text{kg})(9.6\ \text{m/s}^2) - 0.60\ \text{N} = 4.320\ \text{N} - 0.60\ \text{N} = 3.720\ \text{N}$

$\boxed{F_b = 3.72\ \text{N}}$ (the subtraction sets the decimal place; g does not limit the significant figures).

Watch out: keep the sign of a. Using a = 0 just because the indoor helium blimp moves up, or dropping the thrust, gives the wrong buoyant force.

2. Recoil (25 points)



Step 1 **(a) Forces on the tennis-ball launcher while the tennis ball is in the tube.**

Vertical: the ground pushes up (F_N); Earth pulls down (F_g); the tennis ball presses down ($F_{ball,\ down}$). Nothing accelerates vertically, so $F_N = F_g + F_{ball,\ down}$.

Horizontal: the only force is the tennis ball pushing back on the tennis-ball launcher, $F_{ball\ on\ launcher}$, to the left. The vertical forces do not enter the horizontal net-force equation. Arrow lengths are not to scale.

Step 2 **Units and frames.**

Convert first: 57.0 g = 0.0570 kg; 32.6 cm = 0.326 m; 32.0 cm = 0.320 m.

Step 3 **(b) Kinematics for the tennis ball.**

Use the ground-frame pair: the ball's ground distance 0.320 m with its ground speed 16.0 m/s (not the 0.326 m tube length). Time is not given: use #3.

$$v^2 = v_0^2 + 2a\Delta x \rightarrow a_{ball} = \frac{v^2}{2\Delta x} = \frac{(16.0\ \text{m/s})^2}{2(0.320\ \text{m})} = \frac{256\ \text{m}^2/\text{s}^2}{0.640\ \text{m}} = \boxed{4.00 \times 10^2\ \text{m/s}^2}$$

Step 4 **(c) Second law on the tennis ball, then the third law.**

$$a = \frac{F_{net}}{m} \rightarrow F_{launcher\ on\ ball} = m_{ball}a_{ball} = (0.0570\ \text{kg})(4.00 \times 10^2\ \text{m/s}^2) = 22.8\ \text{N} \rightarrow$$

22.8 N, forward on the tennis ball

Third law: $F_{ball\ on\ launcher} = -F_{launcher\ on\ ball}$ → **22.8 N, backward on the tennis-ball launcher**

Step 5 **(d) Second law on the tennis-ball launcher, horizontal only.**

$$a_{launcher} = \frac{F_{ball\ on\ launcher}}{M} = \frac{22.8\ \text{N}}{3.00\ \text{kg}} = \boxed{7.60\ \text{m/s}^2\text{, to the left (backward)}}$$

Check the frames: $t = v/a_{ball} = 0.0400$ s, and the launcher moves $\frac{1}{2}(7.60\ \text{m/s}^2)(0.0400\ \text{s})^2 = 0.0061$ m back. Ball 0.320 m forward plus launcher 0.006 m back = the 0.326 m tube.

Watch out: do not mix frames. The tube length is measured on the launcher; the 16.0 m/s is measured from the ground, so it pairs with the 0.320 m ground distance.

3. Drag linearization (25 points)

Step 1 **(a) Match $F_D = Kv^2$ to $y = mx + b$.**

$F_D = K \cdot (v^2) + 0$

vertical (y)	horizontal (x)	slope	intercept
F_D (N)	v^2 (m^2/s^2)	K	0

Vertical: F_D in N. Horizontal: v^2 in m^2/s^2. Slope = K.

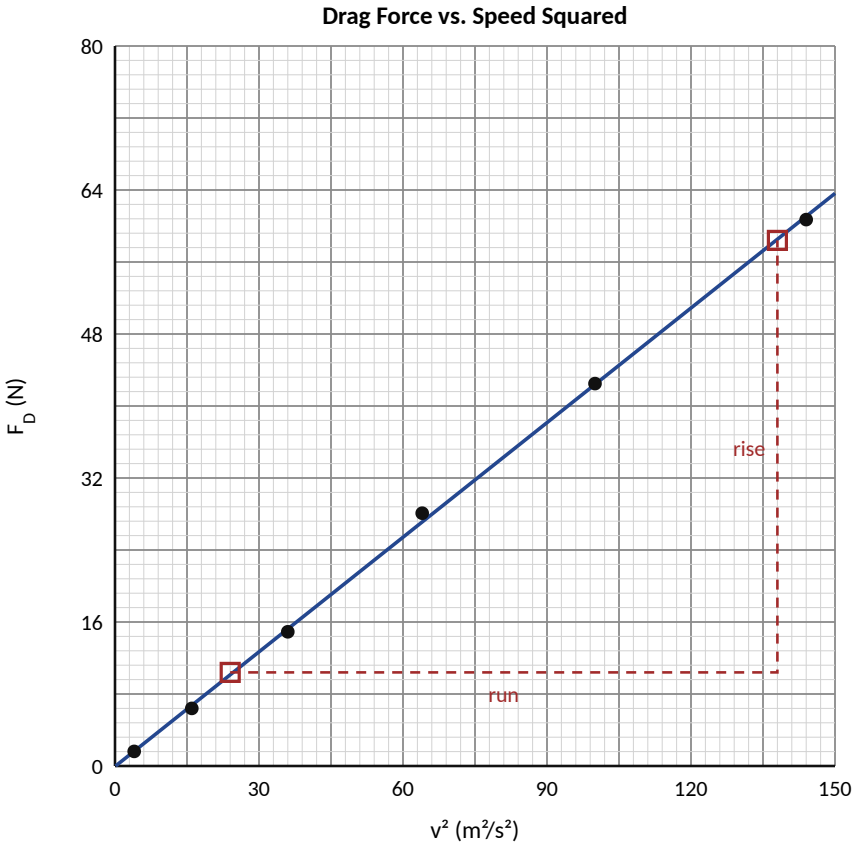
Watch out: v^2 is speed squared, in m^2/s^2. It is not an acceleration (m/s^2).

Step 2 **(b) Square each speed; the units square too.**

Example: $(2.00\ \text{m/s})^2 = 4.00\ \text{m}^2/\text{s}^2$

v (m/s)	F_D (N)	v^2 (m^2/s^2)
2.00	1.62	4.00
4.00	6.40	16.0
6.00	14.9	36.0
8.00	28.1	64.0
10.0	42.5	100
12.0	60.7	144

Step 3 **(c) Graph: labeled axes with units, a scale that uses more than half the grid, one straight line through the middle of the points.**



The best-fit line is not forced through any data point. The red squares are points read off the line, not data points.

Step 4 **(d) Slope from two well-separated points on the line, units carried.**

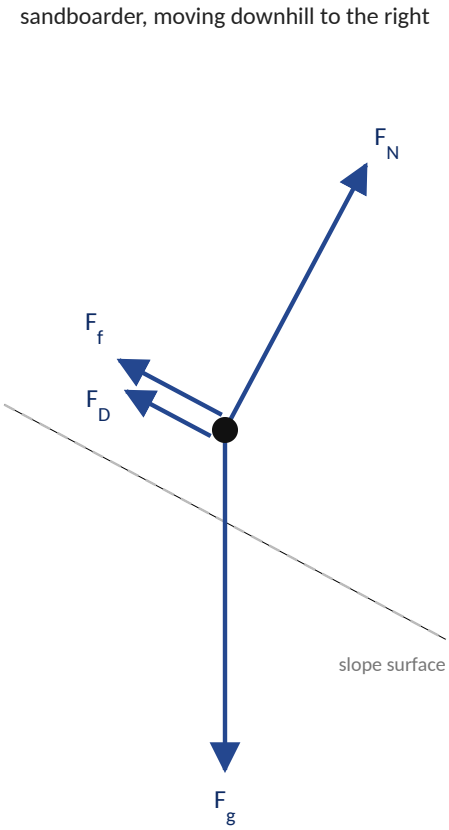
Points read off the line, away from data points: (24 m^2/s^2, 10.4 N) and (138 m^2/s^2, 58.4 N)

$$\text{slope} = \frac{58.4\ \text{N} - 10.4\ \text{N}}{138\ \text{m}^2/\text{s}^2 - 24\ \text{m}^2/\text{s}^2} = \frac{48.0\ \text{N}}{114\ \text{m}^2/\text{s}^2} = 0.42105\ \text{N}\cdot\text{s}^2/\text{m}^2$$

$K \approx 0.421\ \text{N}\cdot\text{s}^2/\text{m}^2\ (= \text{kg/m})$. Any value from 0.391 to 0.459 $\text{N}\cdot\text{s}^2/\text{m}^2$ from a reasonable line earns credit.

Units: $\text{N} \div (\text{m}^2/\text{s}^2) = \text{N}\cdot\text{s}^2/\text{m}^2$, and $\text{N} = \text{kg}\cdot\text{m}/\text{s}^2$, so $\text{N}\cdot\text{s}^2/\text{m}^2 = \text{kg/m}$.

4. Free-body diagram on a slope (25 points)



Step 1 **List every force on the sandboarder: exactly four.**

Earth pulls on it: F_g. The sand pushes on it two ways: F_N (perpendicular to the surface) and kinetic friction F_f (along the surface). The air pushes on it: drag F_D.

Step 2 **Directions. No angle is needed.**

F_g: straight down, vertical, not perpendicular to the slope.

F_N: perpendicular to the slope, pointing out of the surface, so it tilts away from vertical toward the right.

F_f: along the slope, uphill (up and to the left), opposite the sliding.

F_D: opposite the motion through the air. The air is still and the sandboarder moves downhill, so drag also points uphill, parallel to F_f.

Step 3 **Check the diagram.**

Four arrows, each starting on the dot, each labeled. F_f and F_D are two separate arrows, side by side, both uphill. Speeding up, slowing down or moving at a constant speed changes the arrow lengths, not their directions.

Lengths in this key: slowing down, so the two uphill arrows together are longer than the part of gravity that pulls along the slope, and FN is a little shorter than Fg. Lengths are not graded.

Watch out: do not add the components of F_g (along and perpendicular to the slope) as extra arrows; they are not new forces. Drag is the air resistance, not a fifth force, and there is no forward "motion" force.

1. Thrust and buoyancy

Suggested partial credit	pts
(a) F_g down, thrust and buoyancy up, labeled	4
(a) No normal force	2
Acceleration: correct kinematics equation, magnitude and direction (sign)	6
$F_b + F_t - mg = ma$ written with up as +	4
Solved in letters: $F_b = m(g + a) - F_t$	3
Substitution with units in every parenthesis (and grams to kg if needed)	3
$F_b = 3.72$ N	3
Total	**25**

2. Recoil

Suggested partial credit	pts
(a) F_N up, F_g down (the tennis ball pressing down may also be shown); backward push from the tennis ball; no forward force	5
Unit conversions (or none needed) and the matching ground-frame values	3
(b) Kinematics equation chosen and solved in letters; $a_{ball} = 4.00 \times 10^2$ m/s²	5
(c) $F = m_{ball}a_{ball} = 22.8$ N, and the third law gives the same size back on the tennis-ball launcher	5
(d) acceleration = F/M = 7.60 m/s², backward	7
Total	**25**

3. Drag linearization

Suggested partial credit	pts
(a) F_D (N) on the vertical axis, v^2 (m²/s²) on the horizontal axis	4
(a) Slope = K	2
(b) v^2 column correct, with units	4
(c) Axes labeled with quantity and units	2
(c) Scale uses more than half the grid	2
(c) Points plotted correctly	2
(c) One straight best-fit line	2
(d) Two well-separated points read off the line, not data points	3
(d) K from 0.391 to 0.459, with units N·s²/m² (or kg/m)	4
Total	**25**

4. Free-body diagram on a slope

Suggested partial credit	pts
F_g straight down (vertical), labeled	5
F_N perpendicular to the slope, out of the surface, labeled	5
F_f along the slope, uphill, labeled	5
F_D uphill (opposite the motion through still air), labeled, separate from F_f	5
Every arrow starts on the dot; no components and no extra forces	5
Total	**25**